

Cast Vote Records and Ranked-Choice Counting

A congressional election recounted from the ballots, and what the ballots say about the campaigns

84-355 Democracy's Data

August 2026

The instruction at the top of Alaska's ballot is an offer: **rank as many or as few candidates as you like**. Four candidates were listed for the state's single seat in the U.S. House in November 2024, and a voter could put a number beside one of them, or two, or all four.

121,602 Alaskans filled in exactly one oval and left the rest of the grid blank. That was the single commonest piece of paper in the election. The second commonest was the same thing with a different name marked.

We know this because Alaska publishes its ballots.

01 The closest thing to a public record of individual votes

Elections publish **totals**, because the ballot is secret and a total is what is left once the ballots are put away. Alaska does not put them away.

Since adopting ranked-choice voting, the state publishes its **cast vote record**: one row per scanned ballot, showing every ranking on it. No name, no address — but the actual choices, one voter at a time. Nothing else in American election data comes this close to the individual act.

The contest here is the 2024 race for Alaska's at-large House seat: **Mary Peltola**, the Democratic incumbent, against **Nick Begich**, with two minor candidates. The question is what people did with the offer at the top of the ballot.

It is worth asking because **ranked-choice voting is a claim about what a ballot can express**, and the system is spreading. Maine, Alaska, and dozens of cities use it, with repeal fights running in several of them. The claim is that you can support your real first choice and still have a say between the finalists, so a candidate must build a coalition rather than mobilize a faction. Whether that works is an empirical question about what voters write down, and until states published cast vote records it could not be answered at all. A total cannot tell you whether anybody's second preference mattered.

The count itself also becomes contestable in a new way. Under plurality, counting is addition. Under instant-runoff it is a procedure with rules about exhausted ballots, skipped ranks and overvotes. Those rules live in a document rather than in the data.

RANKED CHOICE VOTING

Instructions: Vote by ranking candidates of your choice.

- Rank as many or as few candidates as you like
- Completely fill in no more than one oval for each candidate or column
- For your 1st choice, fill in the oval in the 1st choice column
- For your 2nd choice, fill in the oval in the 2nd choice column
- For your 3rd choice, fill in the oval in the 3rd choice column, and so on
- If you make a mistake, you can ask for a new ballot**

United States Representative					
	1st Choice	2nd Choice	3rd Choice	4th Choice	5th Choice
Duke Caboom (Registered Democrat)	☐	☐	☐	☐	☐
Bob Gunderson (Registered Republican)	☐	☐	☐	☐	☐
Buzz Lightyear (Registered Republican)	☐	☐	☐	☐	☐
Cal Weathers (Nonpartisan)	☐	☐	☐	☐	☐
Write-in:	☐	☐	☐	☐	☐

Figure 1. The instrument: Alaska’s own demonstration ranked-choice ballot. Candidates down the side, ranking columns across; the 2024 contest listed 4 candidates and allowed 4 rankings. The instructions at the top are the ones every Alaskan read before marking anything, and the first of them is the offer this chapter is about. Every row of the file below is one of these grids, flattened. State of Alaska, Division of Elections, retrieved 10 August 2026.

02 One row is one grid

Ballots	1st ranking	2nd ranking	3rd ranking
121,602	Begich, Nick		
86,278	Peltola, Mary S.		
17,294	Peltola, Mary S.	Hafner, Eric	
12,205	Begich, Nick	Howe, John Wayne	
10,674	Peltola, Mary S.	Hafner, Eric	Howe, John Wayne

One row is one **ranking sequence** and how many ballots carried it. The most common in Alaska was **"Begich, and nobody else."** The second most common was **"Peltola, and nobody else."**

A row of that table is a grid with some ovals filled. Here are two of them, beside the record each becomes: the commonest ballot in the election, and the commonest one that used every column.

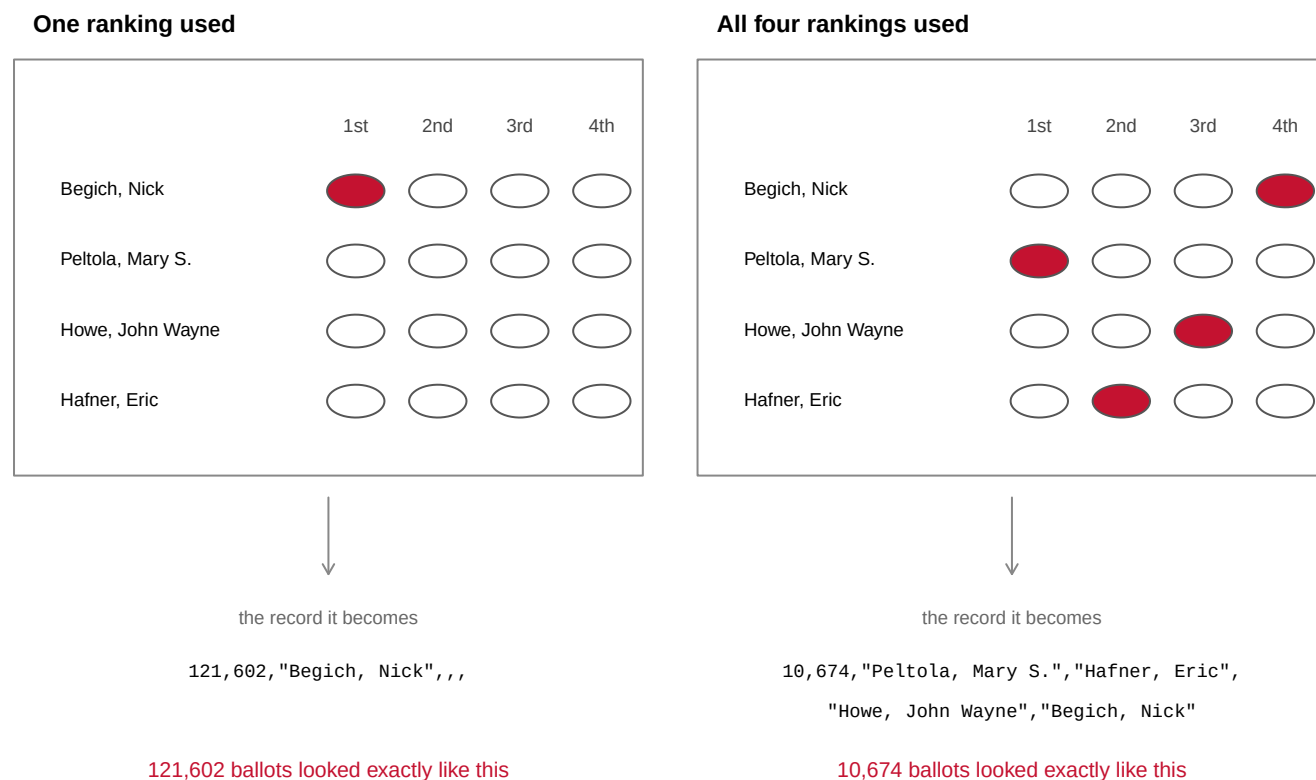


Figure 2. Two ballots, and the two records they become. On the left, the commonest marked grid in the election: one ranking used, 121,602 ballots identical to it. On the right, the commonest grid that used all 4 columns, cast 10,674 times. Underneath each is the row it becomes in the file — a count, then one candidate name per ranking, with an empty field for every column the voter left blank. Everything a cast vote record knows about a voter is in those few characters.

The file behind this chapter has been collapsed one step further than the state's. Precinct and tabulator identifiers were dropped and identical ballots grouped, because a distinctive ranking in a small precinct can narrow down who cast it. **A record with no name on it is not the same as a record nobody can identify.**

03 A file that exists because one state decided it should

Nothing obliged Alaska to publish this. No federal law requires a state to release its ballots, and almost no state does. What the Division of Elections put out after the 2024 general election is the export its voting equipment writes anyway. It is a Dominion JSON dump, one session per scanned ballot, with marks nested inside contests nested inside cards. That had always been an internal by-product of the tabulator. It is a public record now because somebody decided the transparency was worth the trouble.

The scale is worth stating even though it cannot be checked from anything shipped here: the build script records an export of roughly 1.6 GB across a couple of thousand files. Beside it sits a second source of a wholly different character. The official round-by-round result, against which this chapter's own count is checked, is a **PDF**, and somebody had to read it and type its numbers into a table.

It is the appropriate source because the question is about individuals and nothing else

in American elections is. What Alaskans did with the offer at the top of the ballot is not a property of a precinct or a county. It is a fact about single pieces of paper, and a total has already destroyed it.

The improvement that would matter is not a better file but more of them: other states, other years, other contests. Everything below is one race in one state that adopted this system by initiative, and the voters in it are not a sample of anything. The improvement that would matter most is impossible on purpose. To know *why* two-thirds of these ballots used one ranking you would need the voter attached to the ballot, and the secret ballot is the deliberate guarantee that no such file will ever exist.

Two challenges, and the second is the one the chapter has to live with. The first is that this file is not the certified count, and Alaska says so. Ballots counted by hand are not in the export, and write-in names are not shown. So the 340,778 records here are not this contest's turnout.

Nor do the ballots count themselves. The rules that turn these rows into a winner live in a tabulation manual rather than in any column. They cover skipped ranks, two marks in one column, and a ballot that runs out of continuing candidates. That is why a reconstruction from the ballots and the state's own total do not land on precisely the same number.

The second challenge is the deeper one. A ballot is a record of a choice and contains nothing about the reason for it. So the central question of this chapter is not one this record can answer: whether a single ranking is confusion or calculation. It can only be narrowed, by looking at which voters left the other columns blank, and the narrowing is the argument.

The cleaning was done for privacy rather than for tidiness. The build is in Python instead of base R for a plain reason: base R has no JSON reader, and the source is JSON. What it writes is 926 rows standing in for 340,778 ballots, with the precinct and tabulator identifiers removed before anything reached disk. That collapse loses nothing for any question asked here, and cannot be undone for the question of who cast what, which is the point of doing it.

It is worth noticing how unusual that is. Most of the cleaning in this book is repair. This is a deliberate destruction of information, done by the people teaching from the file rather than by the state that released it. A reader rebuilding this would refetch the export, re-run the Python, and retype the official results out of the PDF, which is the only part of the pipeline no script can do for them.

The two ways in cost different things. The export is a public download at a stable address, so it can be fetched again and the count re-run against it. The official rounds were typed by a person out of a document. They cannot be fetched again, cannot be compared automatically, and carry no trail back to the page they came from.

That matters more here than it usually would. A mistyped digit in those tables would not announce itself as a typo. It would show up as a gap between Alaska's count and this chapter's reconstruction, which is precisely the quantity the tables were transcribed in order to measure.

04 What one ballot looks like when a scanner writes it down

The export itself is too large to ship here, but one ballot out of it is not. A single session is kept verbatim from the download, with four fields replaced by the word `MASKED`: the tabulator, the batch, the record number and the image path. They are masked for the same reason the built file drops them, and nothing else on the record has been touched.

Field	Value
TabulatorId	MASKED
BatchId	MASKED
RecordId	MASKED
CountingGroupId	1
ImageMask	MASKED
SessionType	ScannedVote
VotingSessionIdentifier	
UniqueVotingIdentifier	
Original.PrecinctPortionId	MASKED
Original.BallotTypeId	37
Original.IsCurrent	true
...	13 contests on this ballot, one after another

That is one session of 2,028 ballot files, 300 lines long.

There are 13 contests on that ballot and this chapter wants one of them. Here it is, complete, exactly as it appears inside the record above.

Field	Value
Id	7
ManifestationId	662
Undervotes	0
Overvotes	0
Marks.CandidateId	517
Marks.ManifestationId	5017
Marks.PartyId	15
Marks.Rank	1
Marks.MarkDensity	98
Marks.IsAmbiguous	false
Marks.IsVote	true
Marks.OutstackConditionIds	[]
CandidateId	552
ManifestationId	5023

Field	Value
PartyId	7
Rank	2
MarkDensity	92
IsAmbiguous	false
IsVote	true
OutstackConditionIds	[]
CandidateId	520
ManifestationId	5025
PartyId	3
Rank	3
MarkDensity	92
IsAmbiguous	false
IsVote	true
OutstackConditionIds	[]

That is a voter ranking three candidates, and the file does not contain the name of any of them. `CandidateId` is a number whose meaning lives in a different file in the same zip, `CandidateManifest.json`, which lists 209 candidates across every contest on the ballot. Contest 7 is likewise a number, resolved in a third file. Nothing about the record is readable without its dictionaries.

Rank	CandidateId	Who that is
1	517	Begich, Nick
2	552	Hafner, Eric
3	520	Howe, John Wayne

And that is the row. The same ballot, in the file this chapter reads, is one line among 926:

Ballots	1st ranking	2nd ranking	3rd ranking
593	Begich, Nick	Hafner, Eric	Howe, John Wayne

Six judgments stand between the two, and every one of them is a rule somebody chose.

Take the contest, and leave the rest of the ballot alone. A session carries 13 contests; 12 of them are discarded on sight. The whole apparatus is per-ballot, so the decision to study one race is made by a single equality test buried in a loop.

Believe `IsVote`, not the ink. Each mark also carries `MarkDensity`, how dark the oval was, and `IsAmbiguous`. Those are facts about a piece of paper going through a scanner. `IsVote` is the machine's verdict on them, and the build takes the verdict rather than second-guessing it with a density threshold of its own. A different threshold would be a different election.

Group by rank, and decide what two marks in one column mean. Ranks are not positions in a list; they are numbered columns on a grid, and a voter can fill two ovals in the same column. The build treats that as an overvote at that rank and writes `OVERVOTE` in the cell. Nothing in the file says to do this. The state's own tabulator has its own rule, and the difference between the two rules is part of the gap between this chapter's count and Alaska's.

Sort the ranks and close the gaps. A voter who marks a first and a third choice, leaving the second blank, becomes a two-long sequence here. Alaska's tabulator applies a further rule about two or more skipped ranks that this reconstruction does not. That is the honest reason the two counts differ by about a tenth of a percent rather than not at all.

Translate the numbers into names. 517 becomes Begich, Nick by a lookup into a file that is not the data. If the manifest and the ballots ever came from different exports, every ranking in this chapter would be attributed to the wrong candidate and the totals would still be correct.

Then destroy what is left. Ballots with identical sequences are collapsed into one row with a count, and the identifiers never reach disk. That takes 2,028 files and 1.58 GB down to 926 rows, and the compression is not the point of it. The point is that the operation cannot be run backwards. Every question in this chapter survives the collapse; the question of who cast which ballot does not, and that is why it was done.

05 The whole election in 926 rows

Quantity	Value
Ballots in the cast vote record	340,778
Distinct ranking sequences any Alaskan produced	926
Ballots per distinct sequence, on average	368
Sequences used by exactly one ballot	363

340,778 ballots collapse to 926 distinct patterns. 4 candidates and 4 ranks allow far more combinations than that; Alaskans used 926 of them, and most used one of the first few.

06 Two-thirds declined the offer

Candidates ranked	Ballots	% of all ballots
0	10,150	3.0
1	215,501	63.2
2	48,026	14.1
3	18,995	5.6
4	40,409	11.9

Candidates ranked	Ballots	% of all ballots
5	7,697	2.3

Ignore the 10,150 ballots that left this contest blank entirely. Of the 330,628 that voted in it, **215,501 ranked exactly one candidate, or 65.2%.**

Given 4 rankings, two-thirds of Alaskans used one. Neither the advocates nor the opponents of ranked-choice voting advertise that number. The system's entire promise is that you can express a second preference without wasting your vote, and most voters did not.

The obvious explanation is that the system is confusing. It is worth holding that thought, because the file can test it, and the test comes out the other way.

First, though, the count itself.

07 Counting a congressional election from the ballots

Instant-runoff is short enough to state in a sentence. Count everyone's top **continuing** choice. If nobody has a majority, eliminate the lowest and look at those ballots' next choice. Repeat.

Round	Continuing ballots	Leader	Leader's %	Eliminated after this round
1	330,018	Begich, Nick	48.39	Write-in
2	329,614	Begich, Nick	48.49	Hafner, Eric
3	328,300	Begich, Nick	48.76	Howe, John Wayne
4	322,000	Begich, Nick	51.22	—

Begich, Nick wins with 51.22%. That is a congressional election recounted from the ballots themselves, by a rule that fits in three sentences.

Now check it against the state.

Round	Candidate	Official votes	Official %
Round 3	Begich, Nick	164,861	51.22
Round 3	Peltola, Mary S.	156,985	48.78

Alaska's official final round: Begich, Nick 51.22%, Peltola, Mary S. 48.78%. Ours matched to two decimal places.

Quantity	Value
Ballots continuing in our final round	322,000
Ballots continuing in the state's final round	321,846
Difference	0

Quantity	Value
As a share of the count	0.000%

The **percentages** match exactly. The **counts** do not: ours carry 0 more ballots into the final round.

That gap is not a bug to fix. It is the point. The state's tabulator applies rules this recount does not, printed on the first page of its own report. It excludes unresolved write-ins, treats a ballot as exhausted when two or more ranks are skipped, and handles overvotes a specific way.

The rules are not in the data. They are in a separate document, and they change the count. Here by 0.000%. In a closer race, by the winner.

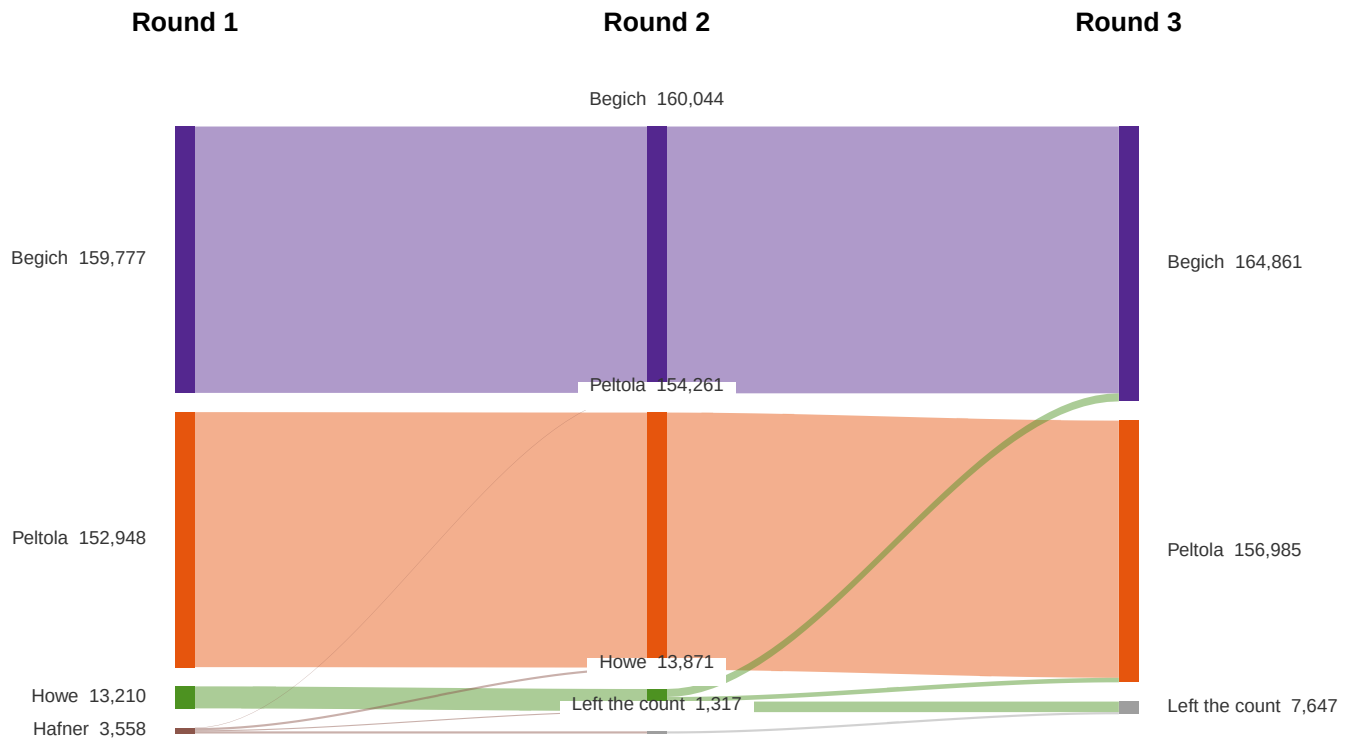
08 Where the eliminated candidates' votes went

Eliminated candidate	Transferred to	Ballots
Hafner, Eric	Begich, Nick	267
Hafner, Eric	Howe, John Wayne	661
Hafner, Eric	Peltola, Mary S.	1,313
Hafner, Eric	Exhausted	1,300
Hafner, Eric	Overvotes	17
Howe, John Wayne	Begich, Nick	4,817
Howe, John Wayne	Peltola, Mary S.	2,724
Howe, John Wayne	Exhausted	6,295
Howe, John Wayne	Overvotes	35

When John Wayne Howe was eliminated, his 13,871 ballots split — and the largest single destination was **nowhere**. 6,295 of them, **45.4%, were exhausted**: those voters had ranked nobody else, so their ballots left the count before the decisive round.

That is the trade at the heart of ranked-choice voting. It lets you express more than one preference. It does not make you.

The whole count is three columns and a set of pipes. Watch the gray one.



Alaska's official rounds, all to the same scale. The gray band is 7,647 ballots (2.3% of 329,493) that ran out of rankings: 7,595 exhausted plus 52 overvotes.

Figure 3. Alaska's official count, all three rounds drawn to one scale. Each column is a round; each band is ballots moving from one round to the next. The gray band is the ballots that leave: 7,647 of the 329,493 that started, 2.3% of the count, made up of 7,595 exhausted ballots and 52 overvotes. Because every band is drawn to the same scale, the width of the gray one is the honest size of the thing the argument about ranked-choice voting is mostly about.

09 Two things only individual records can show

First ranking	Second ranking	Ballots
Peltola, Mary S.	Begich, Nick	10,771
Begich, Nick	Peltola, Mary S.	6,663

The crossover. 17,434 ballots ranked one major candidate first and the other second. That is 5.3% of everyone who voted in this race. Ranked-choice voting is supposed to reward candidates who can be somebody's second choice across the divide. One voter in twenty took the offer.

Candidate named at both rank 1 and rank 2	Ballots
Begich, Nick	2,455
Peltola, Mary S.	1,049

Candidate named at both rank 1 and rank 2	Ballots
Howe, John Wayne	147
OVERVOTE	90

And 3,790 ballots name the same candidate twice. Nothing stopped them: the scanner records the marks it sees, and a voter who wants to be very sure can fill the same column twice. The tabulator skips the duplicate and moves on, so these ballots count normally — but the behavior is in the record, and **no total would ever show it.**

Which brings the argument back to the two-thirds. 65.2% of voters ranked one candidate and stopped. If that is confusion, meaning people did not understand the offer at the top of the ballot, then the rate should be roughly the same whichever candidate they marked. Confusion has no reason to prefer a front-runner.

Before you read on, commit to an answer. The next table splits that 65.2% by who the voter put first. Do Begich's voters and Hafner's voters look alike? Write down two things: whether the gap between them is small — within a few points, as confusion would give you — or large; and if large, whose supporters stopped after one ranking more often. Then turn the page.

10 Who stops after one

First choice	Voters	% who ranked only that candidate
Begich, Nick	159,703	76.1
Peltola, Mary S.	152,887	56.4
Howe, John Wayne	13,146	43.4
Hafner, Eric	3,537	35.4

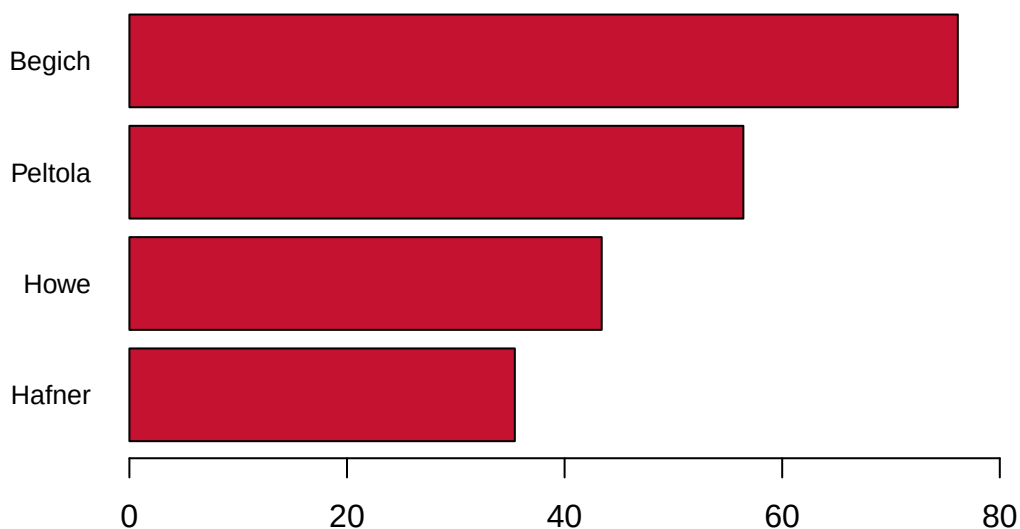
They do not look alike at all. Most readers guess the two shares land within a few points of each other, because that is what a confusing ballot would produce. The gap is 41 points wide, and it runs the opposite way from the guess people make when they assume the smallest candidate's supporters were the most lost.

76.1% of Begich's voters ranked nobody else. 35.4% of Hafner's did. And the pattern runs cleanly down the ballot: the bigger your candidate, the likelier you are to stop after one.

That is not confusion. It is arithmetic that voters appear to have done. If your candidate is the front-runner, a second ranking has almost no chance of ever being read. It only matters if your first choice is eliminated, and your first choice is not going to be eliminated. Ranking nobody else costs you nothing. If your candidate is polling at one percent, a second ranking is the only part of your ballot that will actually do anything.

There is a second reading, and this file cannot separate it from the first: **campaign instruction.** "Rank one and stop" messaging is a documented tactic in RCV races, and a twenty-point gap between the two major candidates' supporters is exactly what it would

produce. Either way, the conclusion is the same and it is the one the totals hide: **these ballots are showing you the campaigns' strategies, not just the voters' preferences.**



% of that candidate's voters who ranked nobody else
Candidates are ordered by how many first-preference votes they received: 159,703 down to 3,537.

Figure 4. Bullet voting is not spread evenly across candidates. The share of each candidate's first-preference voters who ranked nobody else, with candidates ordered by how many first preferences they received — 159,703 down to 3,537. The rate falls from 76.1% among Begich's voters to 35.4% among Hafner's. The browser version toggles to the second view: how many rankings were used across all 340,778 ballots.

11 What the majority is a majority of

Quantity	Value
Ballots in the cast vote record	340,778
Continuing in our first round	330,018
Continuing in the final round	322,000
Winner's final-round total	164,935
As a share of continuing ballots	51.22%
As a share of all ballots cast	48.4%

The winner's 51.22% is a majority of the **322,000 ballots still in the count**, not of the 340,778 people who cast one. Measured against every ballot, the figure is 48.4% — a plurality.

This is the live objection to ranked-choice voting, and it has a literature. Burnett and Kogan (2015) examined four ranked-choice elections and found that a large share of ballots were exhausted before the final round. So the eventual winner's "majority" was a majority of a shrunken pool rather than of the people who voted.

Critics of the system treat that as disenfranchisement by arithmetic. **The majority is partly manufactured by shrinking the denominator**, the bottom number the winner's votes are divided by. The figures above are what that looks like in one race.

The fair answer is a question: what would the alternative denominator mean? Under plurality the winner's share here would also have been under half, and there would have been no second round for anyone's ballot to survive into. A voter who declines to rank anyone else has made a choice. Donovan, Tolbert and Gracey (2016, 2019) find that voters in ranked-choice cities report more civil campaigns and generally say they understand the system. That cuts against the reading that these were people who did not follow the instructions. Figure 4 is the sharper version of the same point: the people who stopped after one ranking were disproportionately the ones with least to gain from a second.

12 What the ballots can testify to

Three arguments run through this material, and it is worth separating what the ballots settle from what they only frame.

The first is whether publishing ballots is safe at all. A cast vote record carries no name and is a public record. It is also individual-level, and a sufficiently unusual ranking in a small precinct narrows down who cast it. That is the real objection. It is why the file behind this chapter has no precinct column and no individual ballots in it: 340,778 records were collapsed to 926 sequences with counts before anything was saved.

Alaska decided the transparency was worth it. Most states have decided nothing, because most states publish nothing.

The second is exhaustion, and this race puts a number on it rather than resolving it. It matters arithmetically — 7,647 ballots, 2.3% of the count, left before the decisive round. Whether it matters democratically depends on what you think a ballot is for, and that is not a question a file of ballots can answer.

The third is what one race can settle, which is very little. Begich won under instant-runoff and also led on first preferences, so the mechanism did not change the winner here. It changed the *claim*, from a plurality to a majority of continuing ballots. Anyone using this contest to argue for or against ranked-choice voting in general is generalizing from one case.

The evidence itself is unusually strong in one respect and narrow in several. It is individual-level, complete for the ballots it covers, and published by the state that ran the election. It also reproduced the official result to two decimal places, which is a stronger check than almost anything else in this field.

But it is not the certified count. Alaska states plainly that the cast vote record excludes ballots counted by hand, so 340,778 is not the official turnout for this contest. The tabulation rules that turn these rows into a winner live in a separate document and move the count by 0 ballots. A ballot records a choice and nothing about why, so there are no demographics and no reasons in any of this. And it is one race, one state, one year.

What survives is worth having. Given 4 rankings, 65.2% of Alaskan voters used one; 5.3% crossed between the two major candidates; 3,790 ballots named the same person twice. None of that is visible in any published total. Bullet voting is not uniform and is not plausibly confusion, running from 76.1% among the front-runner's voters to 35.4% among the

smallest candidate's — which is what strategic voters, or well-instructed ones, would produce. And the ballots do not count themselves.

Two questions are left genuinely open, and more ballots would not close either. Whether bullet voting reflects a calculation voters made or an instruction they were given: the pattern is rational either way, and separating the two needs survey data these rows cannot supply. And whether ranked-choice voting changes who runs and how they campaign, which is a claim about candidates rather than about ballots.

What the ballots do show, finally, is not only preferences. A twenty-point gap between two major candidates' supporters in how many ovals they filled is a fact about campaigns as much as about voters. The grid at the top of this chapter offered every Alaskan as many or as few rankings as they liked, and the answers came back sorted by whose interest it was to use them.

13 Sources

Data

- State of Alaska, Division of Elections, 2024 General Election **cast vote record**, published as a Dominion JSON export. <https://www.elections.alaska.gov/results/24GENR/>
- Official round-by-round results and transfer detail from the state's **RCV Detailed Report**, <https://www.elections.alaska.gov/results/24GENR/RCV-USRep.pdf>
- The ballot in Figure 1 is the State of Alaska's own **sample ranked-choice ballot**, cropped to the ranked-choice instructions and the U.S. Representative contest. <https://www.elections.alaska.gov/doc/GenRCVsampleBallot11.9.21.pdf> Retrieved 10 August 2026 and reproduced here, so *reading* this document needs no network. *Building* it does: the ranked-ballot figure pulls d3 from d3js.org at knit time, and pandoc embeds that copy in the HTML it writes.
- The committed file contains no individual ballots: 340,778 records were collapsed to 926 ranking sequences with counts, and precinct and tabulator identifiers were dropped, before anything was saved.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`official_rounds.csv`, `official_transfers.csv`, `sequences.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `sequences.csv` has one row per distinct ranking, with a `ballots` count. Count the ballots that ranked only one candidate. What does that say about how well the ballot instructions worked?
- Find the sequences in `sequences.csv` that rank candidates in an order nobody campaigned for. Which pairs of candidates share voters?
- `official_transfers.csv` records where each eliminated candidate's ballots went. Compare that against what you would guess from the parties involved. Were any transfers surprising?

- `official_rounds.csv` shows the count at each stage. Work out who led in the first round and who won. Then decide whether the first-round leader has a complaint.

Academic literature

- Burnett, C. M. and Kogan, V. (2015). "Ballot (and Voter)'Exhaustion' Under Instant Runoff Voting: An Examination of Four Ranked-Choice Elections." *Electoral Studies* 37: 41-49.
- Donovan, T., Tolbert, C. and Gracey, K. (2016). "Campaign Civility Under Preferential and Plurality Voting." *Electoral Studies* 42: 157-163.
- Donovan, T., Tolbert, C. and Gracey, K. (2019). "Self-Reported Understanding of Ranked-Choice Voting." *Social Science Quarterly* 100(5): 1768-1776.

How the figures were made

The state's Dominion JSON export is read once. The individual ballots are collapsed to 926 ranking sequences with counts, and the precinct and tabulator identifiers are dropped. What is written out is `sequences.csv`, alongside the official round and transfer tables typed up from the state's own report. The instant-runoff count above is run in this document, from those sequences, at the moment it is rendered; so is every number in every figure and caption.

REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

I want to rebuild a public dataset from its original source, without any starting files. Please do the following and show your work.

THE SOURCE. State of Alaska, Division of Elections, cast vote record export for the 2024 general election:

https://www.elections.alaska.gov/results/24GENR/CVR_Export_20241130154411.zip

No account or key is needed. The zip is about 44 MB and expands to about 1.58 GB of JSON. Inside, one "session" per scanned ballot, each carrying cards, contests, and marks; every mark names a candidate and a rank. The U.S. Representative contest carries the Id 7, and a manifest file inside the zip maps candidate Ids to names. This is the closest thing to a public record of individual votes that exists in the United States. The state's own round-by-round result for the same race is a small PDF beside it, `RCV-USRep.pdf`, on the same site. Two warnings from Alaska itself: ballots counted by hand are not in the export, and write-in names are not shown, so the file will not reproduce the certified totals exactly – and is not supposed to.

HOW TO FETCH IT. The download itself is plain. The size is the trap: read the JSON files one at a time from inside the zip rather than expanding all of it into memory at once.

WHAT TO BUILD.

1. Every ballot's ranking in the U.S. Representative race, collapsed to distinct ranking sequences with a count of how many ballots cast each. Nothing identifying any voter survives the collapse.
2. Your own instant-runoff count from those sequences, eliminating the last-place candidate each round, next to the state's official round-by-round result.

CHECK YOUR WORK. The copy this book used was fetched in August 2026.

If your rebuild matches it, you should find:

- 2,043 JSON files of ballots inside the archive
 - 340,778 ballots rank someone in this race, collapsing to 926 distinct ranking sequences
 - the official final round: Begich 164,861 votes to Peltola's 156,985 – 51.22 against 48.78 percent
 - your own count reproduces those percentages exactly, while the raw vote counts sit about 150 ballots apart in 322,000; the official tabulator applies exhaustion rules the simple count does not
- This is an archived election, so the file should not change; the timestamp in its name would change if the state ever reposted it.